

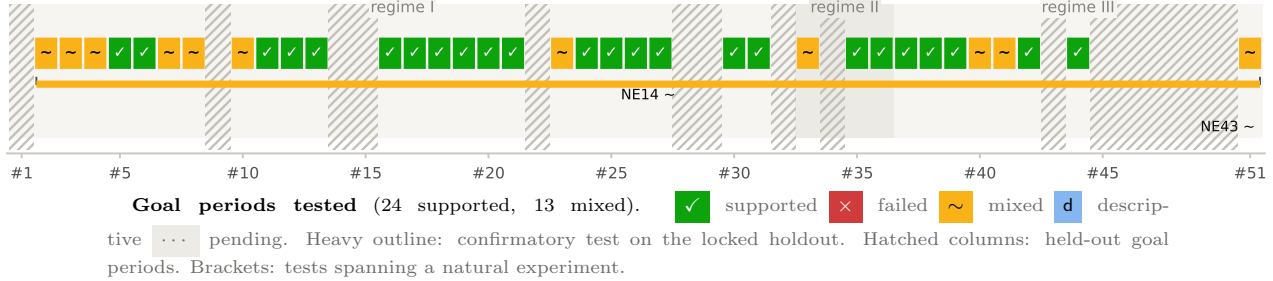
H50: Field or coupling: transfer functions and lag tell which

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-05

Complete 60% Faithfulness 2.5/5 Usefulness 2.5/5

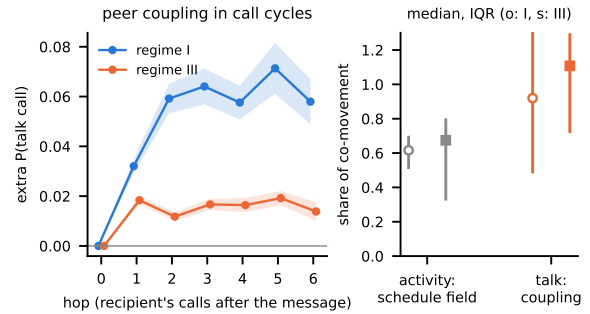
Bottom line. Activity co-movement is a field set by the scheduler; talk and content spread by a coupling gated at one read-out call, and in the always-on regime only to named agents.

level: hypothesis · pipeline checks 86% · updated 2026-10-05



WHAT HAPPENED

The question is whether agents co-move because a common input hits everyone (a field) or because they read each other (a coupling). In call cycles, a field moves every agent at its next call. A coupling acts only at the recipient's first call that can read the message. The DQ1 context ledger gives a placebo-corrected discontinuity, hop by hop: calls that start just after a message against calls just before it. The analysis also fits input impulse responses and decomposes co-movement on 71 non-holdout units (35 periods). **Activity co-movement is the schedule:** day start and stop explain about 0.6 of it beyond a null. **Talk co-movement is a gated coupling.** A peer message raises the chance that the next call is a talk call (45/71 units, none negative), always at hop 1. It accounts for roughly all talk co-movement. HH167's regime split fails. **Round 2: content couples at the read-out call too** (regime III, matched age: $J_f^c = 0.033 [0.027, 0.039]$, named $\times 5$). Graph-clause lags stay unresolved: the pre-registered event study is length-biased, and reads batch. In regime I the talk jump is mostly a switch into chat mode. The holdout is not run.



Observables. Left: excess talk probability of a recipient's call, by hop after a peer message (IVW pooled; 95% band). The response is zero at the in-flight call (hop 0) and steps up at the read-out call (hop 1). Regime III (13-s chained calls) drops at hop 2, then plateaus. Regime I rises to hop 3; round 2 finds the rise not significant on logged-start recipients and mostly a switch into chat mode. Right: share of activity co-movement explained by the schedule edges (gray), and of talk co-movement by the gated coupling (orange; > 1 is overshoot).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not read synchrony in activity as influence. The scheduler's daily start and stop explain about 0.6 of activity co-movement; agents do not cause it. Influence acts at the recipient's next call and turns a work call into a talk call. In the always-on regime it needs the recipient's name: the jump is 0.17 for named recipients and 0.004 for others. To propagate a message, address the agents by name and expect a response within one call (about 15–25 s; minutes if the agent pauses). The nudger moves single agents and leaves the collective mode unchanged. Bookend messages are announcements, not inputs. To move what an agent says, name it: content pull is about 5 times larger for named messages.

PHYSICAL MODEL

Gated linear response (models 02, 01). Agent i makes calls c at times t_c . A message posted at t_m enters the context at the first call with $t_c > t_m$ (hop 1). The call outcome Y_c (talk, work, idle) responds to inputs u_{c-k} (bookends, human messages, nudges, platform errors). It also responds to peer messages m_{c-k} first read k hops earlier. s is the call start minus the message time. e_{it} is agent-day-demeaned minute activity or talk. S is the equal-time co-movement.

$$\text{logit } P(Y_c=1) = h_i + \sum_k g(k) u_{c-k} + \sum_k J(k) m_{c-k}$$

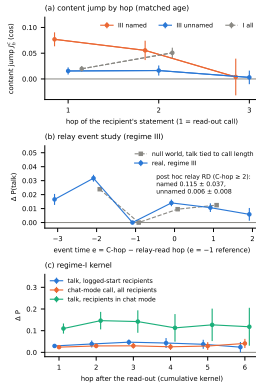
$$J_k = E[Y | s \rightarrow 0^+] - E[Y | s \rightarrow 0^-] - \text{placebo}, \quad \delta(h) = \sum_{k \leq h} J_k$$

$$S = \sum_t [(\sum_i e_{it})^2 - \sum_i e_{it}^2] / \sum_t \sum_i e_{it}^2$$

H50-field-vs-coupling-transfer-lag · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H50-field-vs-coupling-transfer-lag/README.md · ratings: H50 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 synthetic recovery	J_1 bias -2 to -17% (III); dead time $2 \rightarrow$ onset hop 3; nulls 3/35	passed
P2 inputs at read-out	in-room human $J_1 > 0$, hop 1; nudged agents talk at hop 1 (0.14), not later	mixed
P3 low-pass, corner 5–40 min	coherence $8-10\times$ at 15–60 min; mixed corner ill-defined (undershoot)	
P4 field: zero relative lag	onsets IQR 9–22 s (< 2 cycles); supported #38 agents halted at pause	
P5 gated coupling, hop 1	$J_1 > 0$ in 45/71, < 0 in 0; onset hop 1; no decay in regime I	mostly
P6 decomposition	activity field 0.55 (III) $\checkmark$, 0.51 (I, pred. ≤ 0.2) $\times$; talk coupling $> 57/68$	mostly
HH167 regime split	regime-I calls not message-triggered (ratio 1.03–1.07)	rejected
NE43 switch-offs	nudge path gone; coupling $\times 0.69$; edge field persists	mixed
G51 human relay	plain human messages: hop-1 field; relay underpowered	mixed
G38 pause gates	82% start before any peer talks; supported 94% never call after pause	
R1 content gate (III)	$J_1^c = 0.033$ bge, 0.046 gte; supported 10/16 units; named 0.077 vs 0.016	
R6 graph clause	event study length-biased; $\rho=1$ in 60%; post hoc relay RD 0.014 (named 0.115)	inconcl.
R2 regime-I kernel	no latency artifact; rise n.s. mixed (withdrawn); chat-mode switch $\approx 70\%$ of jump	mixed



Second figure. Round 2. (a) Content jump by hop, statements made at hop h minus in-flight statements at matched age (95% CI): regime III couples at hop 1, named messages $\times 5$. (b) The relay event study (blue) has a pre-trend at $e = -2$; a no-coupling world where talk depends on call length (gray) reproduces it. (c) Regime-I cumulative kernel: logged-start recipients (blue), the chat-mode channel (orange) and recipients already in chat mode (green).

GOAL PERIODS IN DETAIL

G02 $\sim$ mixed
G03 $\sim$ mixed
G04 $\sim$ mixed
G05 $\checkmark$ supported
G06 $\checkmark$ supported
G07 $\sim$ mixed
G08 $\sim$ mixed
G10 $\sim$ mixed
G11 $\checkmark$ supported
G12 $\checkmark$ supported

+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	2	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Regime-I call starts are latency-placed. The jump survives on logged-start recipients (#24–#31; 0.030 vs 0.034), but early regime I is unchecked. Beyond hop 1 the regime-I kernel mixes coupling with the chat-mode sequence (round 2). Per-unit CIs are mildly anti-conservative (synthetic false positives about 9%). Decomposition shares are model-based and do not add up, because coupling amplifies fields. $f_C > 1$ is overshoot at low talk co-movement. Room changes and roster joins confound NE43. The #51 target and relay tests have about 100 pairs. Round 2: two-room weeks have too few in-flight statements for the content gate; the relay RD is post hoc and invalid in regime I; the regime-I kernel shape mixes coupling with the mode sequence. The holdout is not run.

ROUND 2

Is the regime-I switch into chat mode a field or a coupling? Pre-register the relay RD for the reserved data. Test call-indexed designs on length-dependent null worlds. Confirm on the reserved data, including NE23 (nudge off and on).

- **Essence**, which levers move a swarm together (fields) and which only propagate through who reads whom (couplings), measured in the units the scaffold imposes (calls), so an operator knows what an input will do to collective behavior.
- **H50-R3** HH188 frequency-dependent T-eff.
- **H50-R4** HH190 agent impedance.
- **H50-R5** Human-input relay with power.
- **H50-R7** Confirm on the holdout (confirm.py: CP1–CP5, NE23 nudge off/on).
- **H50-R8** Regime-I mode channel: field or coupling?
- **H50-R9** Graph clause, confirmatory.
- **H50-R10** Length-dependent null worlds for every hop-indexed design.
- **H50-R11** Content gate in the two-room weeks (#36–#44) with pooled age bins.